Set It and Forget It - Secure & Automated Certificate Management on AKS

Wolfgang Ofner

Senior Cloud Architect

WARSAW IT DAYS

A lecture selected by a Program Council consisting of recognized leaders in the IT and Data Science field.

Warsaw,
19.03.2026 - 20.03.2026

OFFICIAL LECTURE OF THE WARSAW IT DAYS

Event organized by: Academic Partners Foundation

Agenda

Cert-Manager

Let’s Encrypt

Domain Ownership Validation

Demo

Wolfgang Ofner

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Focus on Azure, Kubernetes, and DevOps

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Microsoft CERTIFIED CYBERSECURITY ARCHITECT EXPERT

Microsoft CERTIFIED AZURE SOLUTIONS ARCHITECT EXPERT

Microsoft CERTIFIED DEVOPS ENGINEER EXPERT

Microsoft CERTIFIED TRAINER MCT

Cert-Manager

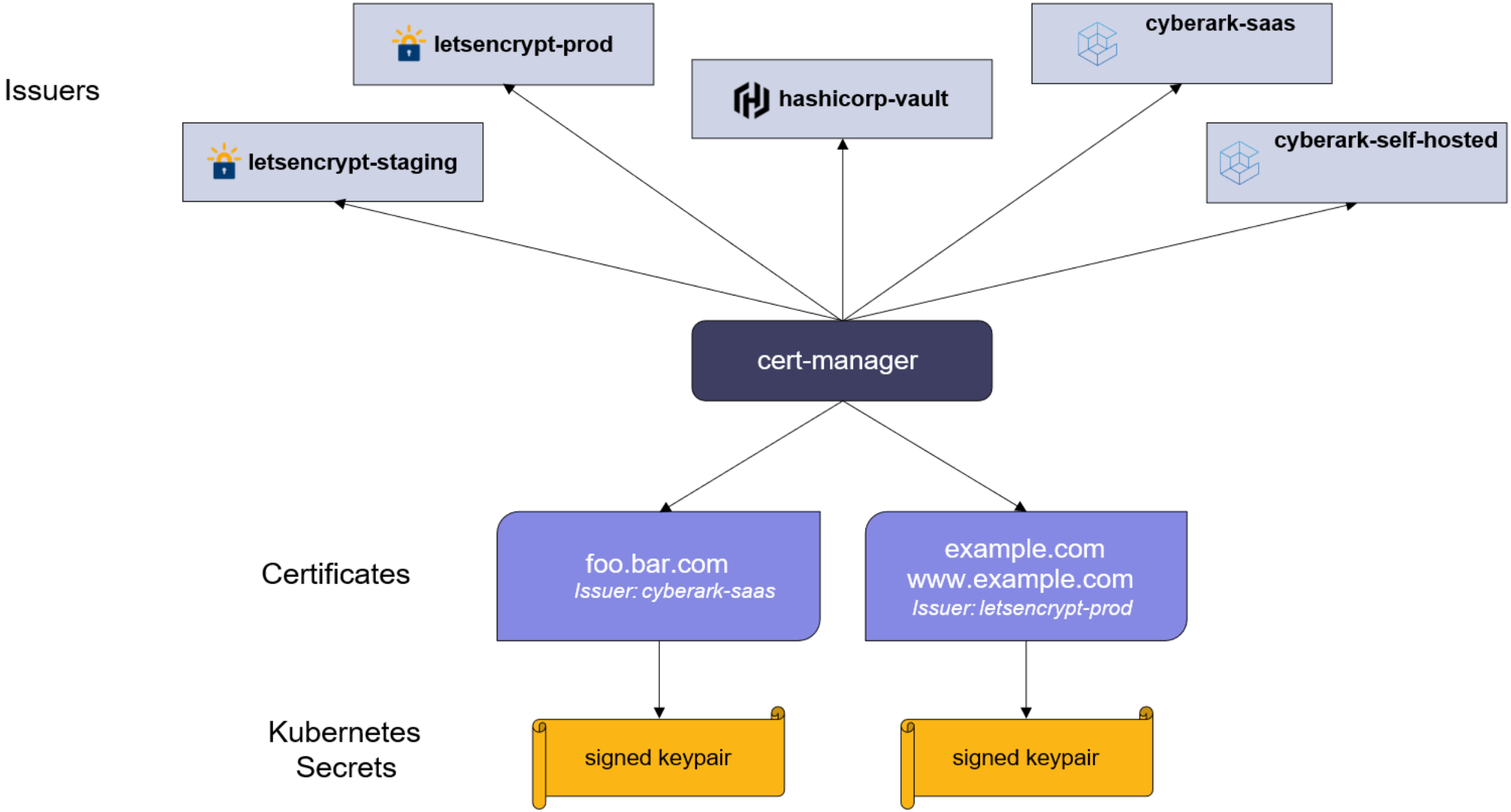
Cert-Manager

Automatically creates/updates certificates

Uses Issuer to get certificates

Supports Ingress and Gateway API

Verifies ownership of Domain


Issuers
letsencrypt-prod
hashicorp-vault
cyberark-saas
letsencrypt-staging
cyberark-self-hosted
cert-manager
Certificates
foo.bar.com
Issuer: cyberark-saas
example.com
www.example.com
Issuer: letsencrypt-prod
Kubernetes Secrets
signed keypair
signed keypair

Let’s Encrypt

Let’s Encrypt

Nonprofit to create a more secure internet

Free certificates, valid for 90 days

Staging and Production environment

Prod certificates are fully valid in modern browsers

Domain Ownership Validation

Domain Ownership Validation

HTTP-01 challenge

```yaml
solvers:
  - http01:
      gatewayHTTPRoute:
        parentRefs:
          - name: $GatewayName
            namespace: $GatewayNamespace
            kind: Gateway
```

```yaml
solvers:
- http01:
    ingress:
      class: nginx
```

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Domain Ownership Validation

HTTP-01 challenge

Proof ownership of a domain with a file

http://<YOUR_DOMAIN>/.well-known/acme-challenge/<TOKEN>

Cannot be used for wildcard certificates

Domain Ownership Validation

DNS-01 challenge

```yaml
- dns01:
    azureDNS:
      hostedZoneName: $DnsZoneName
      resourceGroupName: $DnsResourceGroup
      subscriptionID: $DnsSubscriptionId
      environment: AzurePublicCloud
      managedIdentity:
        clientID: $CertManagerManagedIdentityClientId
```

Domain Ownership Validation

DNS-01 challenge

Proof ownership of a domain with DNS record

_acme-challenge.<YOUR_DOMAIN>

Challenge can be delegated to another DNS zone

Demo

Demo

Deploy new AKS cluster

Use Envoy Gateway API

Deploy Cert-Manager

Create ClusterIssuer with HTTP-01 Challenge

Deploy Demo app to test HTTPs connection

Demo

Create and configure Managed Identity for DNS access

Update Cert-Manager to use Entra Workload ID

DNS-01 Challenge

Test wildcard certificate

Thank you for watching!

Remember to leave your questions
and rate the presentation
in the section below.

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